

COURSE OUTCOME

Students will be able to:

- Design and build web applications using servlets and JSP Manage sessions in servlets and JSP
- Identify where and when to use MVC design pattern Create custom tag in JSP
- Develop web application using struts
- Develop database application using hibernate Develop IOC and DI using springs
- Develop web application using springs.

Unit No.	Title of The Unit	Time required for the Unit (Hours)
1.	Java Servlets	08
2.	Java Server Pages(JSP)	07
3.	Java Server Faces	08
4.	Hibernate	08
5.	Springs	07

A. OUTLINE OF THE COURSE

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B. DETAILED SYLLABUS

Unit	Unit Details
1.	Java Servlets
	• Introduction of Unit • Servlets and HTTP Servlets, Filters, Security, Servlet Life Cycle, Servlets for the World Wide Web, Requests, Responses, and Headers, GET and POST, HTTP, Deploying a Servlet , Web Application Deployment Descriptor Structure, Servlet Configuration, Http Servlet Request/Response, Servlet Context, Session Management, • Case Study • Conclusion of Unit
2.	Java Server Pages(JSP)
	• Introduction of Unit: JavaBeans, Custom Tags and JSP Fragments, JSP Life Cycle, The Difference Between Servlets and JSP, JSP Syntax and Semantics, Elements and Template Data, JSP Configuration, Standard JSP Actions, Attributes, Comments, Quoting and Escape Characters, Exception Handling, JavaBeans and the JSP Expression Language, JSP Standard Tag Library, Custom Tag Libraries, Database Connectivity, Building a Complete Web Application. • Case Study • Conclusion of Unit
3.	Java Server Faces
	• Introduction of Unit: features, life cycle, manage Beans, UI Components- input Text, output Text, form, command Button, input Text Area, input Hidden, input File, Bean, Validation, facelets, JSF JDBC, JSF with controllers, architectural overview of application developed with JSF and JSP, validator tag, data tables. • Conclusion of Unit

4.	Hibernate
	• Introduction of Unit: advantages, features, Architecture, Environment, Life Cycle, ORM Tool, First program, Sessions, Session factory, Persistent Class, Using the Session, MVC, Hibernate Query language, Criteria Query, Mapping Types, Annotations, Query Language, Native SQL. • Case Study • Conclusion of Unit
5.	Springs
	• Introduction of Unit: Architecture, Environment Setup, Create Sample Program, IOC Containers, Bean Definition, Bean Scopes, Bean Lifecycle, Dependency Injection, IOC Injection, Setter Injection, Injecting Inner Beans, Injecting Collection, Event Handling, MVC Framework. • Case Study • Conclusion of Unit

C. RECOMMENDED STUDY MATERIAL

S. No	Text Books:	Author	Edition	Publication
1.	Servlets and Java Server Pages	Jayson Falkner		Kevin Jones-2003
2.	Beginning Hibernate	Joseph B. Ottinger, Jeff Linwood, Dave Minter		Apress-2014
Reference Book				
3.	Professional Java Development with the Spring Framework, Rod Johnson, 8th edition –Wiley			
4.	Core Java Server Faces, David M. Geary, 2004 – 3rd Edition-Prentice Hall			
Online Resources				
5.	https://www.simplilearn.com/resources-to-learn-java-programming-article			
6.	https://www.docdroid.net/mY1yTPu/advancedjavaprogrammingbyuttamkumarroy-pdf			
7.	https://www.edureka.co/blog/advanced-java-tutorial			